

## Project Overview

This project presents the development of an industrial SCADA application created in AVEVA InTouch. The system simulates a liquid mixing and distribution process consisting of three tanks, a conveyor belt, and a product filling station. Operators can monitor process variables, control valves, observe tank levels, and supervise the mixing ratio in real time.

The application also includes a warehouse management module for product classification, carton packaging, and order preparation. The HMI follows Situational Awareness principles and provides a multi-level interface with alarm management, role-based access control, and process monitoring.

## HMI Design

### Dashboard

The Dashboard serves as the primary operator interface, providing a high-level overview of the entire production process. It was designed according to Situational Awareness principles to enable rapid assessment of the system status and early detection of abnormal operating conditions.

W oknie prezentowane są wyłącznie najważniejsze informacje procesowe, takie jak:

- Tank A, B and C levels
- Mixture concentration
- Conveyor status
- Product position
- Active alarms
- Warehouse summary
- Logged-in user
- Current date and time

The Dashboard is intended for monitoring rather than direct process control. It also serves as the default view for users who are not authenticated.

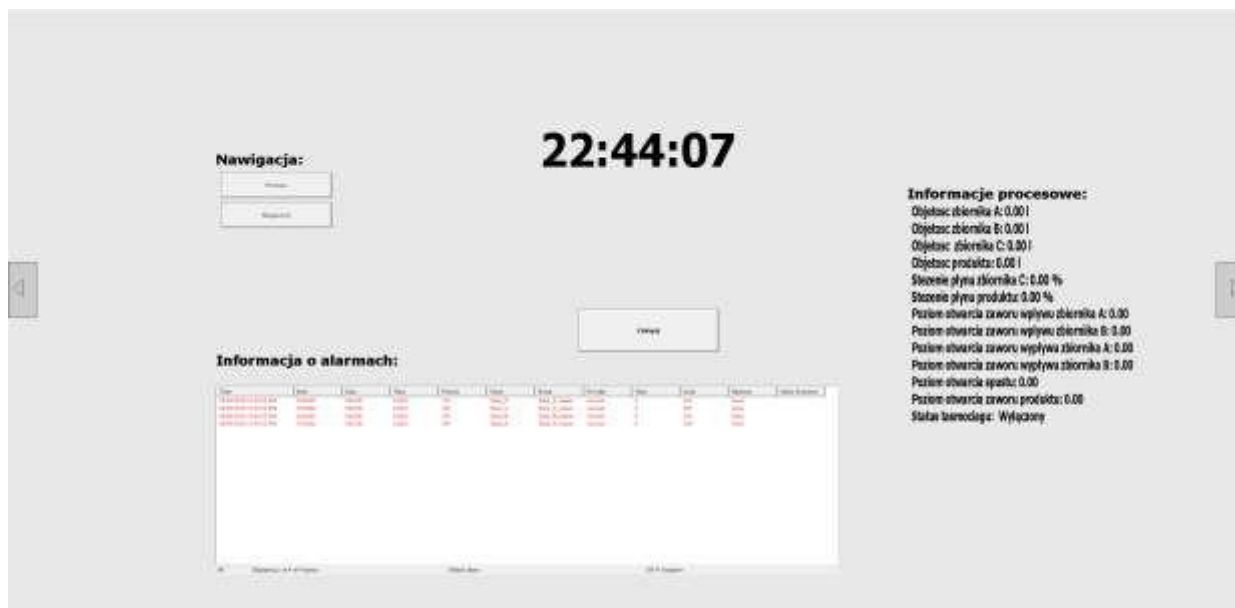


Figure 1. Main SCADA Dashboard

## Process Window

The Process Window provides detailed visualization and manual control of the technological process. Operators can control inlet and outlet valves, monitor tank levels, observe the conveyor operation, and supervise the product filling process.

Real-time process values, including tank volume, liquid concentration, and product fill level, are continuously displayed to support process operation.

The Process Window also supports semi-automatic conveyor operation, allowing operators to monitor the container position using photoelectric sensors and control conveyor movement. During the filling process, the operator can monitor the container fill level and the mixture concentration in real time. The window is intended for continuous process supervision and operational control.

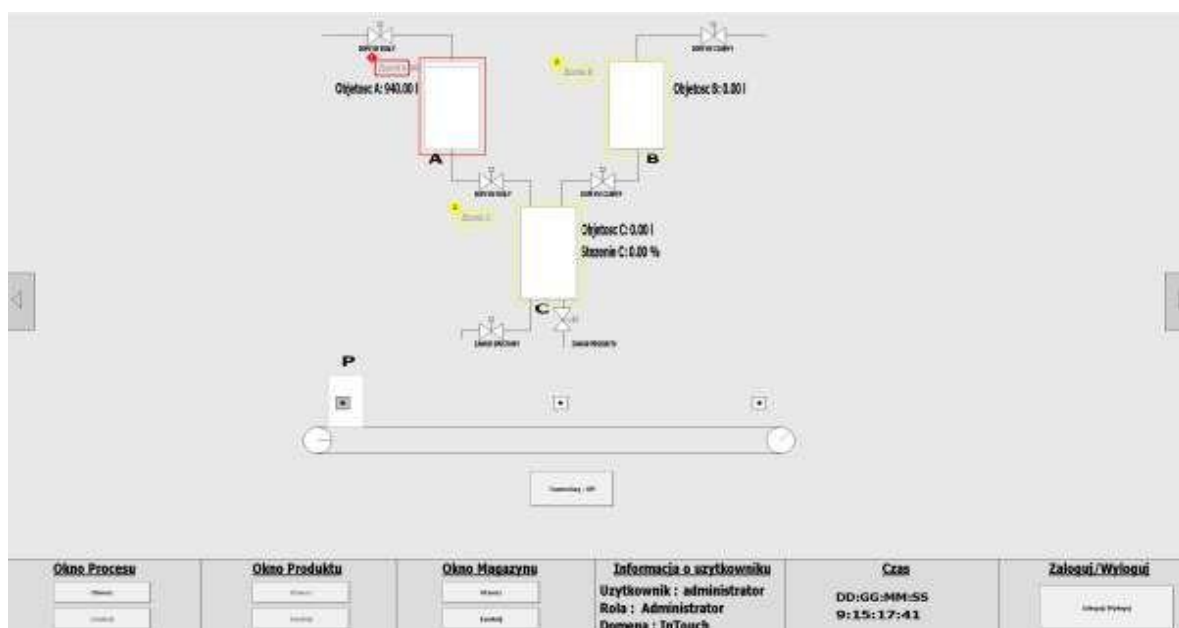


Figure 2. Process Window

## Warehouse Window

The **Warehouse Window** is used to monitor finished products and prepare shipment orders. Products are automatically classified into five quality categories based on the mixture concentration.

Operators can select the number of cartons to be shipped for each product category using dedicated controls. The application verifies warehouse availability and limits each shipment to a maximum of **16 cartons**.

The current shipment is summarized in real time, displaying the number of selected cartons for each product category and the total shipment size. A dynamic loading bar visualizes the composition of the shipment and the remaining transport capacity.

The loading bar was implemented using **Animation Links** with the **Object Size** property. Five segments represent individual product categories, while the sixth segment indicates the remaining available transport capacity. The size and position of each segment are updated dynamically according to the current shipment configuration.

When a shipment is confirmed, the selected cartons are automatically removed from the warehouse inventory and prepared for dispatch.

The warehouse module was designed with future integration into **MES** and **ERP** systems through a relational SQL database in accordance with the **ISA-95** architecture.

The screenshot displays the Warehouse Window interface. At the top, there is a list of five product categories with their respective carton counts and a total sum. Below this is a dynamic loading bar with six segments of different colors (white, light gray, medium gray, dark gray, black, and blue). The bottom of the window features a navigation bar with six sections: 'Okno Procesu', 'Okno Produktu', 'Okno Maszyn', 'Informacja o użytkowniku', 'Czas', and 'Zaloguj/Wyloguj'.

| Kolor       | 3 (20) | + | -  | 2 |
|-------------|--------|---|----|---|
| Biały       | 3 (20) | + | -  | 2 |
| Jasnoszary  | 3 (20) | + | -  | 1 |
| Szary       | 3 (20) | + | -  | 2 |
| Ciemnoszary | 3 (20) | + | -  | 3 |
| Czarny      | 3 (20) | + | -  | 2 |
| Suma :      |        |   | 10 |   |

Okno Procesu   Okno Produktu   Okno Maszyn   Informacja o użytkowniku   Czas   Zaloguj/Wyloguj

Użytkownik : administrator  
Rola : Administrator  
Domena : InTouch

00:00:MM:SS  
9:15:21:03

Figure 3. Warehouse Window

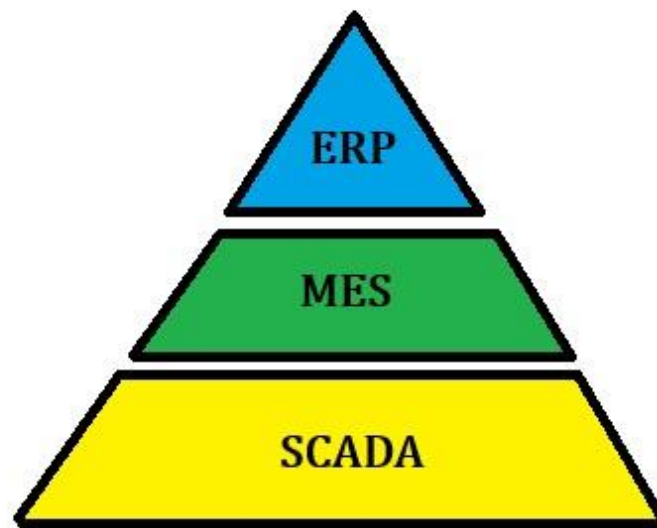


Figure 4. SCADA, MES, and ERP Architecture (ISA-95)

### Tank Detail Windows

Each process tank has a dedicated detail window that provides detailed process information and direct operator control. These windows display key operating parameters, including:

- Current liquid volume
- Tank fill level
- Valve status
- Flow rates
- Active alarm states (HiHi, Hi, Lo, and LoLo)
- Animated tank level visualization

Operators can directly control the inlet and outlet valves according to their assigned access permissions.

The **Tank C** detail window additionally displays the product container during the filling process. It provides real-time information about the container fill level and the mixture concentration while the container is positioned beneath the filling valve.

The purpose of the detail windows is to support process diagnostics, monitoring, and control by providing detailed information about individual process units.

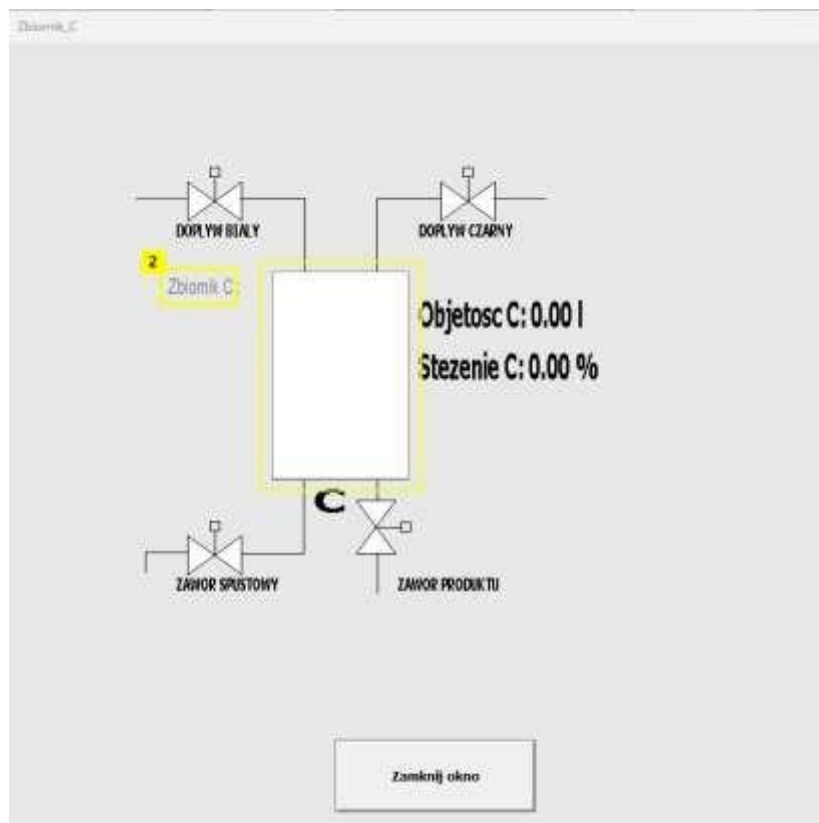


Figure 5. Tank A Detail Window

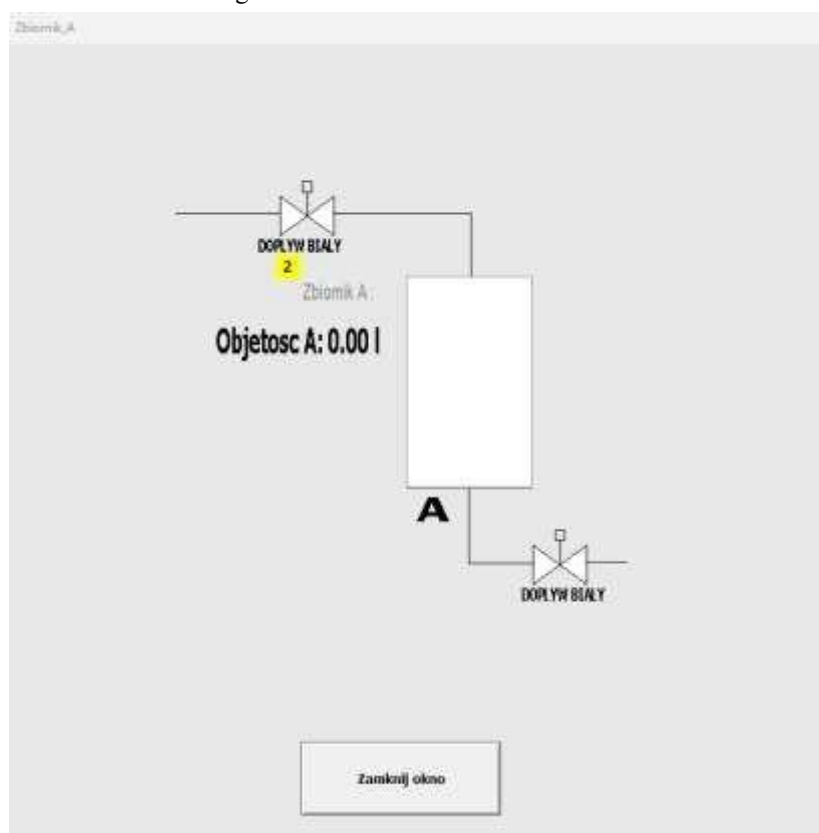


Figure 6. Tank B Detail Window

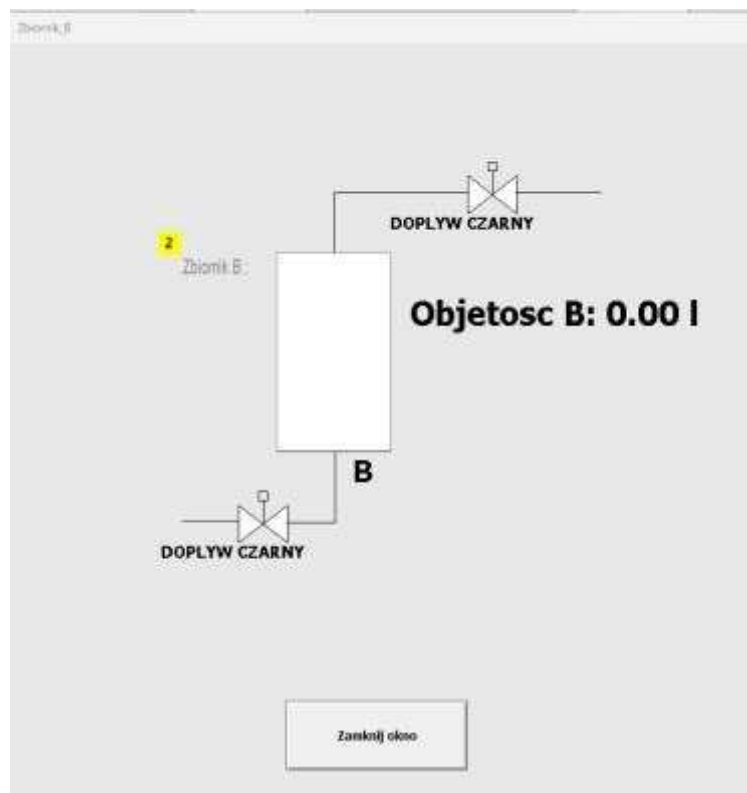


Figure 7. Tank C Detail Window

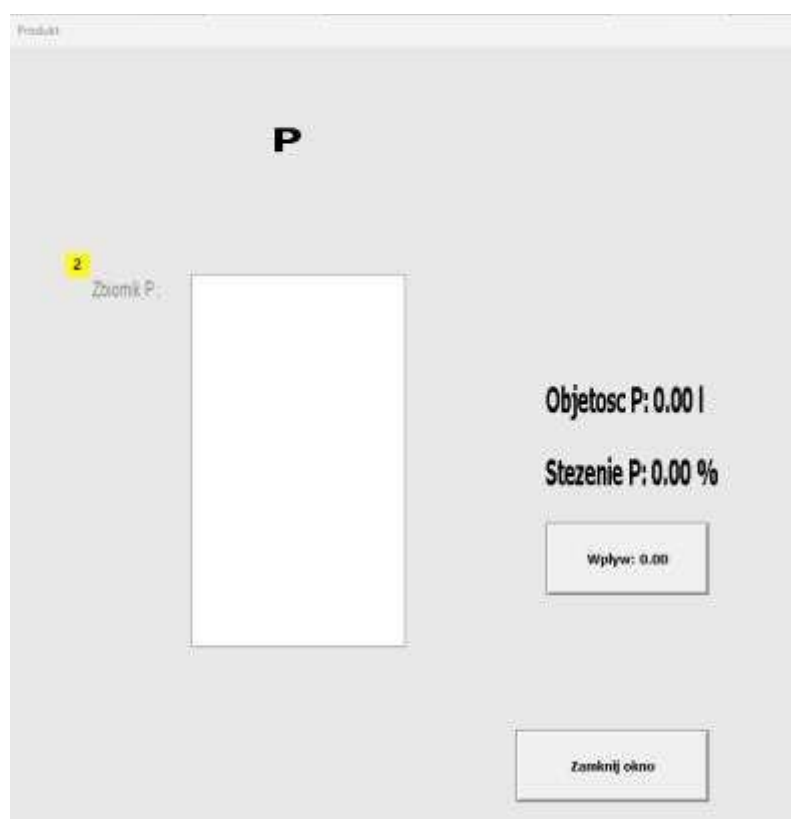


Figure 8. Product Container Detail Window

A dedicated **User Authentication** window was also implemented to support user login and logout operations.



Figure 9. User Authentication Window

### User Access Control

The application implements **role-based access control (RBAC)** to ensure safe operation of the process and restrict access to control and warehouse functions. Each user role is assigned a specific set of permissions based on operational responsibilities.

| Role               | Permissions                                        |
|--------------------|----------------------------------------------------|
| Administrator      | Full system configuration and administration       |
| Tank Operator      | Tank filling operations                            |
| Process Operator   | Complete process control                           |
| Warehouse Operator | Warehouse management and shipment preparation      |
| Supervisor         | Access to all monitoring and operational functions |

## Alarm Management

Alarm groups and alarm conditions were configured for each process tank to improve operator awareness and support fault detection. The HMI provides visual indication of alarm states, including **HiHi**, **Hi**, **Lo**, and **LoLo** level alarms, enabling rapid identification of abnormal operating conditions.

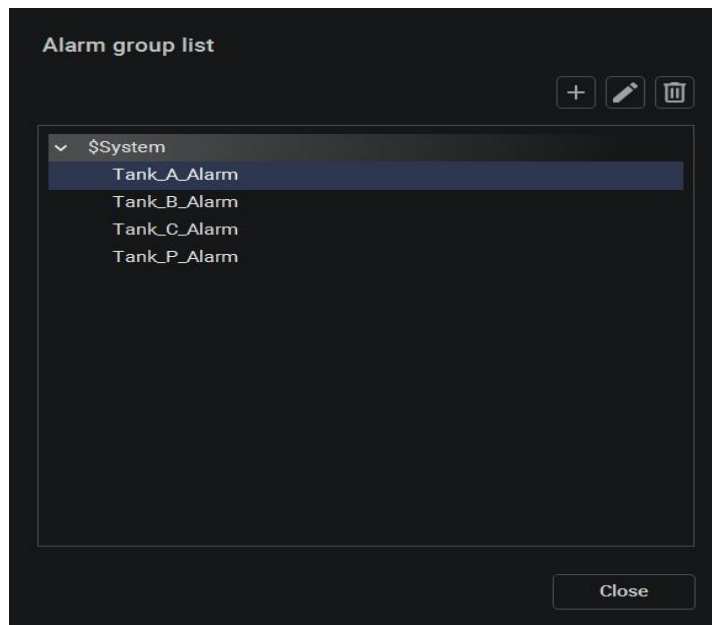


Figure 10. Alarm Group Configuration

The screenshot shows the "Tagname Dictionary" window with the "Details & Alarms" tab selected. The "Tagname" field contains "Tank\_A" and the "Group" field contains "Tank\_A\_Alarm". The "Type" is set to "Memory Real" and "Local Tag" is unchecked. The "Comment" is "AccessLevel". The "Log Data" and "Log Events" checkboxes are unchecked. The "Retentive Value" and "Retentive Parameters" checkboxes are unchecked. The "Initial Value" is 0, "Min Value" is 0, "Deadband" is 0, "Eng Units" is empty, "Max Value" is 1000, and "Log Deadband" is 0. The "ACK Model" is set to "Condition". The "Alarm Comment" field is empty. The "Alarm Value" table is as follows:

|                                          | Alarm Value | Priority | Alarm Inhibitor |                                          | Alarm Value | Priority | Alarm Inhibitor | Value Deadband |
|------------------------------------------|-------------|----------|-----------------|------------------------------------------|-------------|----------|-----------------|----------------|
| <input checked="" type="checkbox"/> LoLo | 100         | 249      |                 | <input checked="" type="checkbox"/> High | 800         | 250      |                 | 0              |
| <input checked="" type="checkbox"/> Low  | 200         | 250      |                 | <input checked="" type="checkbox"/> HiHi | 900         | 249      |                 |                |

The "% Deviation" table is as follows:

|                                          | % Deviation | Target | Priority | Alarm Inhibitor | Deviation Deadband % |
|------------------------------------------|-------------|--------|----------|-----------------|----------------------|
| <input type="checkbox"/> Minor Deviation | 0           | 0      | 1        |                 | 0                    |
| <input type="checkbox"/> Major Deviation | 0           |        | 1        |                 |                      |

The "Rate of Change" section shows "Rate of Change" set to 0, "% per" set to "Min", "Priority" set to 1, and "Alarm Inhibitor" set to empty.

Figure 11. Tank A Alarm Configuration



Tagname Dictionary

☐ Main
 ☐ Details
 ☐ Alarms
 ☒ Details & Alarms
 ☐ Members

New Restore Delete Save << Select... >> Cancel Close

Tagname: Tank\_B Type: ... Memory Real ☐ Local Tag  
 Group: ... Tank\_B\_Alarm ☐ Read only ☒ Read Write  
 Comment: Poziom płynu w zbiorniku B  
☐ Log Data ☐ Log Events ☐ Retentive Value ☐ Retentive Parameters

Initial Value: 0 Min Value: 0 Deadband: 0  
 Eng Units: Max Value: 1000 Log Deadband: 0

ACK Model: ☒ Condition ☐ Event Oriented ☐ Expanded Summary Alarm Comment:

|                                          | Alarm Value | Priority | Alarm Inhibitor |    | Alarm Value                              | Priority | Alarm Inhibitor | Value Deadband |
|------------------------------------------|-------------|----------|-----------------|----|------------------------------------------|----------|-----------------|----------------|
| <input checked="" type="checkbox"/> LoLo | 100         | 250      |                 | .. | <input checked="" type="checkbox"/> High | 800      | 250             | .. 0           |
| <input checked="" type="checkbox"/> Low  | 200         | 500      |                 | .. | <input checked="" type="checkbox"/> HiHi | 900      | 249             | ..             |

|                                          | % Deviation | Target | Priority | Alarm Inhibitor | Deviation Deadband % |
|------------------------------------------|-------------|--------|----------|-----------------|----------------------|
| <input type="checkbox"/> Minor Deviation | 0           | 0      | 1        |                 | 0                    |
| <input type="checkbox"/> Major Deviation | 0           |        | 1        |                 |                      |

☐ Rate of Change 0 % per: ☐ Sec ☒ Min ☐ Hr Priority: 1 Alarm Inhibitor

Figure 12. Tank B Alarm Configuration

Tagname Dictionary

☐ Main
 ☐ Details
 ☐ Alarms
 ☒ Details & Alarms
 ☐ Members

New Restore Delete Save << Select... >> Cancel Close

Tagname: Tank\_C Type: ... Memory Real ☐ Local Tag  
 Group: ... Tank\_C\_Alarm ☐ Read only ☒ Read Write  
 Comment: Poziom płynu w zbiorniku C  
☐ Log Data ☐ Log Events ☐ Retentive Value ☐ Retentive Parameters

Initial Value: 0 Min Value: 0 Deadband: 0  
 Eng Units: Max Value: 3000 Log Deadband: 0

ACK Model: ☒ Condition ☐ Event Oriented ☐ Expanded Summary Alarm Comment:

|                                          | Alarm Value | Priority | Alarm Inhibitor |    | Alarm Value                              | Priority | Alarm Inhibitor | Value Deadband |
|------------------------------------------|-------------|----------|-----------------|----|------------------------------------------|----------|-----------------|----------------|
| <input checked="" type="checkbox"/> LoLo | 300         | 500      |                 | .. | <input checked="" type="checkbox"/> High | 2400     | 250             | .. 0           |
| <input checked="" type="checkbox"/> Low  | 600         | 750      |                 | .. | <input checked="" type="checkbox"/> HiHi | 2700     | 249             | ..             |

|                                          | % Deviation | Target | Priority | Alarm Inhibitor | Deviation Deadband % |
|------------------------------------------|-------------|--------|----------|-----------------|----------------------|
| <input type="checkbox"/> Minor Deviation | 0           | 0      | 1        |                 | 0                    |
| <input type="checkbox"/> Major Deviation | 0           |        | 1        |                 |                      |

☐ Rate of Change 0 % per: ☐ Sec ☒ Min ☐ Hr Priority: 1 Alarm Inhibitor

Figure 13. Tank C Alarm Configuration

## Technical Challenges

The development of the application highlighted several engineering challenges related to process control, system integration, and HMI design.

One of the main challenges was maintaining a correct mass balance during liquid mixing and transfer between tanks. Additional validation logic was implemented to prevent negative flow values, tank overflows, and invalid concentration calculations.

Another challenge was synchronizing the conveyor system with the filling process. The position of the product container had to be accurately coordinated with the photoelectric sensors and the filling valve to ensure reliable operation and prevent simultaneous movement and filling.

The implementation of the warehouse module required additional control logic to validate inventory levels, enforce shipment capacity limits, and automatically classify products according to the mixture concentration. These mechanisms ensured data consistency between the production process and warehouse operations.

The project also demonstrated the importance of designing a multi-level HMI based on Situational Awareness principles. Separating the interface into overview and detail screens improved usability while allowing operators to quickly access detailed process information when required.

Role-based access control was implemented to restrict user permissions according to operational responsibilities, improving system safety and preventing unauthorized actions.

Although some interface elements were created using standard AVEVA InTouch objects instead of ArchestrA Graphics, the final application provides full functionality, reliable process control, and a clear operator interface.